



# **INSURANCE RISK & POLICY ANALYTICS DASHBOARD**

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# 1. Introduction

## 1.1 Purpose

The goal of this design paper is to serve as a strategic plan for constructing a data-driven dashboard using Microsoft Power BI. The dashboard will provide insights into client demographics, risk profiles, and behavioral preferences based on insurance data. This helps insurance businesses make educated decisions relating to policy administration, marketing tactics, and client interaction.

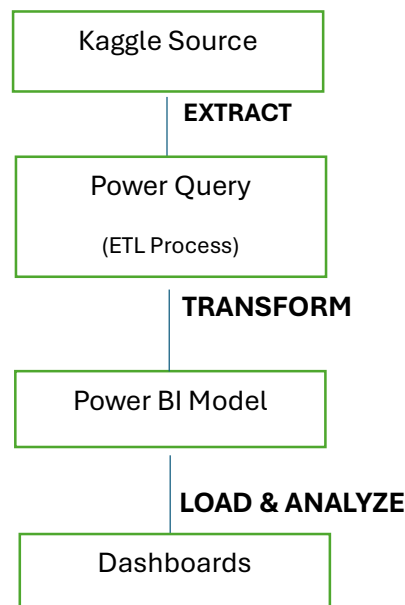
## 1.2 Scope

This study focuses on analyzing a comprehensive insurance dataset to find trends in policyholder behavior, estimate risks, and determine communication preferences. The scope covers data extraction, transformation, modeling, and visualization. Predictive modeling, real-time data intake, and automated warning systems are not within the current scope but can be studied as future expansions.

# 2. System Overview.

## 2.1 System Architecture

The following architecture shows the data flow from source to dashboard:



## 2.2 System Components

User Interface: Power BI dashboards that display KPIs, charts, and slicers

Backend Processing: Performed using Power Query and DAX calculations

Data Storage: In-memory storage within Power BI model

## 3. Data Management

### 3.1 Data Sourcing

Source Name: Insurance Claims and Policy Data

Description: A synthetic dataset comprising customer profiles, insurance policies, claims, communication preferences, and risk classifications.

### 3.2 ETL Process

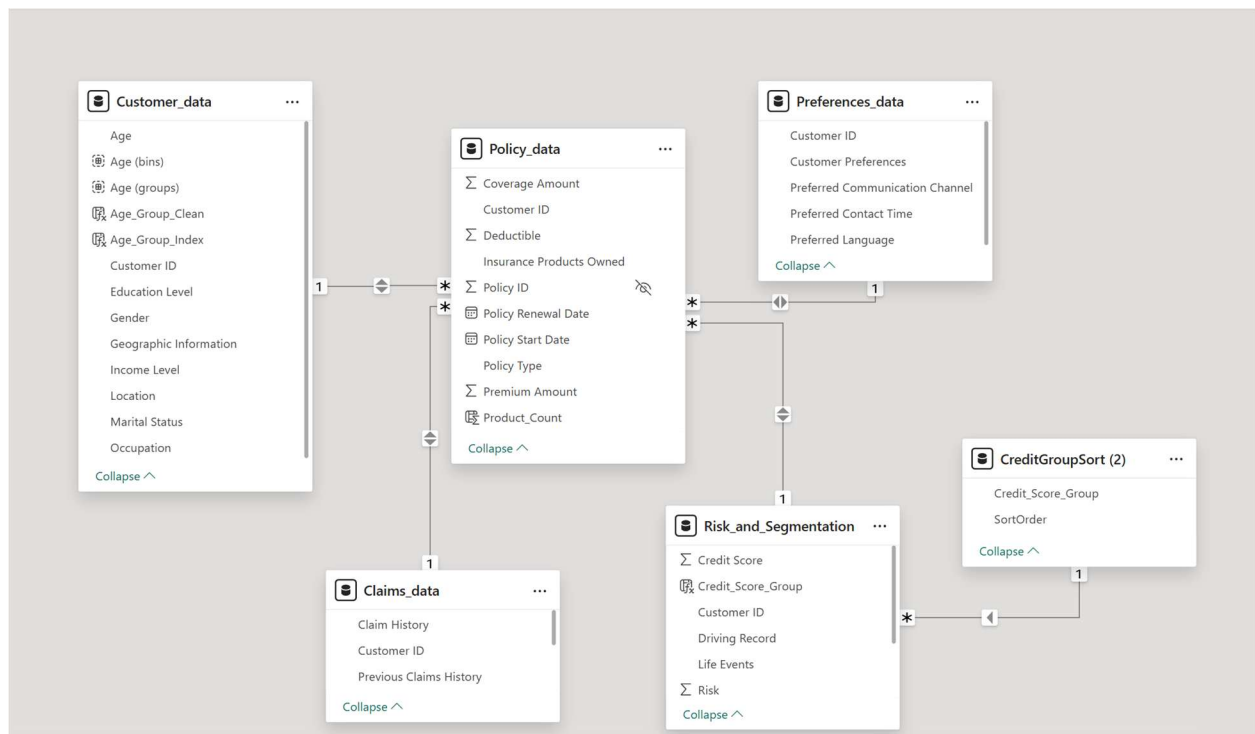
Extract: Imported data from multiple CSV files into Power BI

Transform: Cleaned column headers, standardized formats, Removed duplicates and null values, Created new calculated columns like Product Count, RiskColor, and AgeGroup

Load: Final data tables loaded into Power BI data model using relationships for query optimization

## 4. Data Modelling

### 4.1 Entity-Relationship Diagram



## 4.2 Schema Design

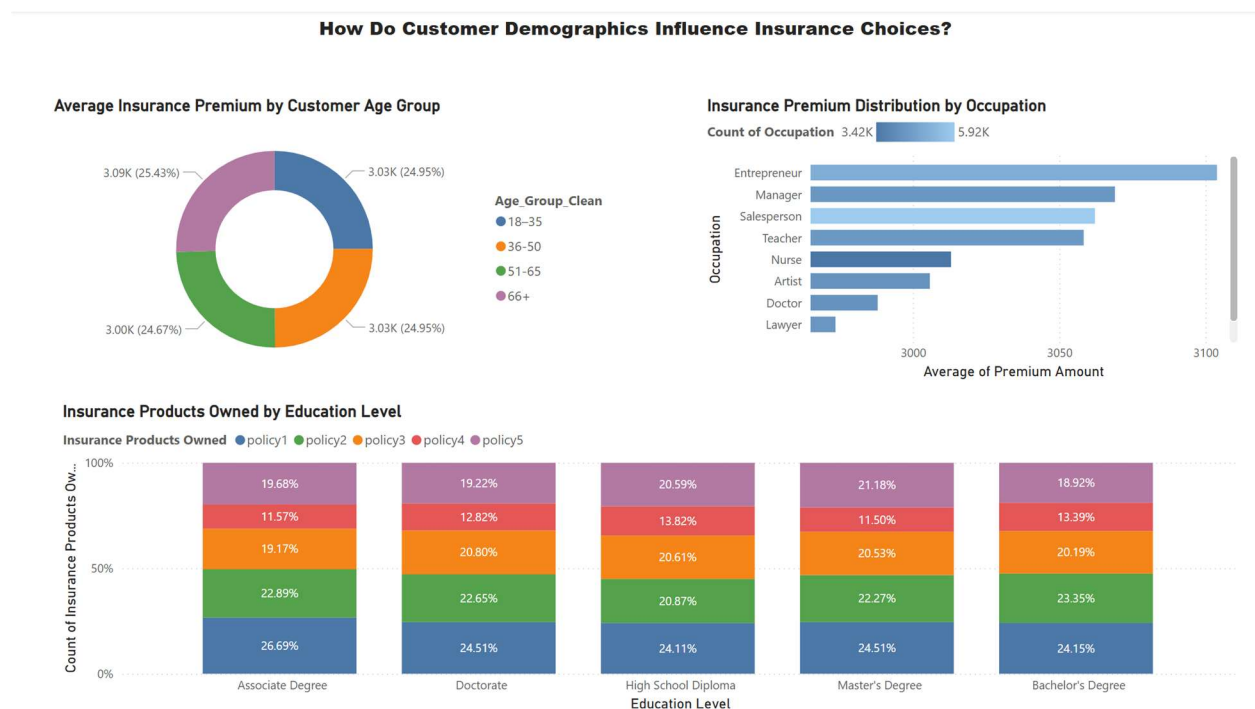
- Customer\_data: Customer\_ID, Age, Gender, Marital Status, Occupation, Education Level
- Policy\_data: Customer\_ID, Policy\_Start\_Date, Premium\_Amount, Coverage\_Amount, Insurance\_Products\_Owned
- Claims\_data: Claim\_ID, Customer\_ID, Claim\_Amount, Claim\_History, Interactions with Customer Service
- Preferences\_data: Customer\_ID, Preferred\_Channel, Contact\_Time, Language\_Preference
- Risk\_and\_Segmentation: Customer\_ID, RiskLabel, Credit\_Score\_Group
- CreditGroupSort: Credit\_Score\_Group, SortOrder

## 5. Visualization & User Interface

### 5.1 Layout and explanation

The dashboard is divided into three thematic sections:

#### 1. Customer Demographics & Premium Trends



**Donut Chart:** Average Insurance Premium by Customer Age Group.

What It Shows: This chart compares average premium amounts paid by customers across different age groups. “This donut chart shows that all age groups pay nearly the same average premium—around 3,000.” “The slight differences suggest age is not a major factor in premium amount, but younger age groups (18–35) and older (66+) are slightly more cautious or limited in investment.”

**Horizontal Bar Chart:** Insurance Premium Distribution by Occupation

What It Shows: Average insurance premium amounts based on different occupations.

“Entrepreneurs pay the highest premiums, likely due to higher income and greater risk coverage needs.” “Doctors and lawyers pay less on average, which could indicate they rely on professional benefits or group plans.” “This chart helps us understand which professions invest more in insurance.”

**3. Stacked Column Chart:** Insurance Products Owned by Education Level

What It Shows: The variety and count of insurance products owned across different education levels

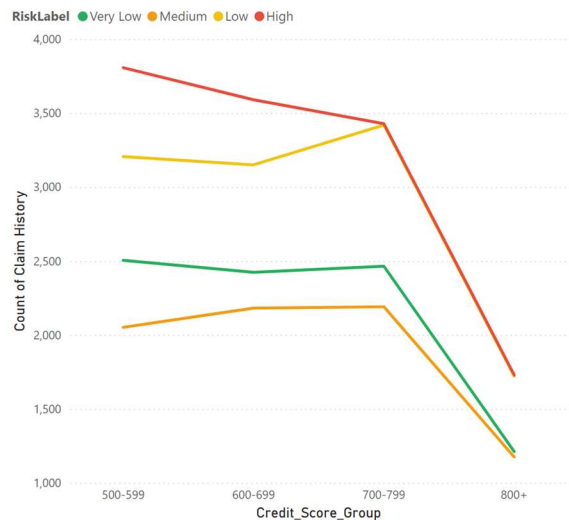
“Customers with higher education levels tend to own a more balanced mix of insurance products.” “Associate degree holders have a higher count for policy1, while master’s and doctoral holders show a more even spread.” “This shows educated individuals likely diversify their insurance portfolio better.”

**2. Risk and Claims Insights**



**Identify how customer risk levels impact insurance claims**

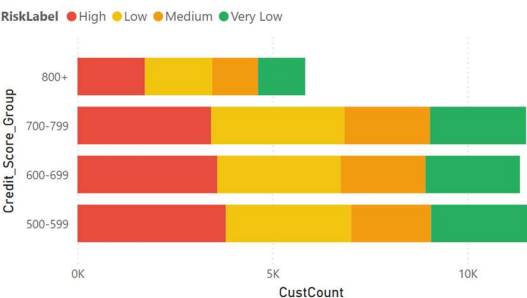
Number of Claims by Credit Score Group and Risk Level



Customer Risk Profile by Life Events

| Life Events | High  | Low   | Medium | Very Low | Total |
|-------------|-------|-------|--------|----------|-------|
| Marriage    | 3416  | 2913  | 1758   | 2076     | 10163 |
| Job Change  | 2570  | 2509  | 1605   | 1903     | 8587  |
| Divorce     | 2210  | 2137  | 1618   | 1675     | 7640  |
| Childbirth  | 2182  | 1947  | 1355   | 1466     | 6950  |
| Retirement  | 2177  | 1991  | 1265   | 1486     | 6919  |
| Total       | 12555 | 11497 | 7601   | 8606     | 40259 |

Distribution of Customer Risk Levels Across Credit Scores



1. **Line Chart:** Number of Claims by Credit Score Group and Risk Level

What It Shows: How the count of claims varies based on both credit score ranges and risk categories (High, Medium, Low, Very Low).

“This chart shows that customers with lower credit scores and higher risk levels file more claims.” “As credit scores increase, the number of claims generally decreases, especially for High-Risk customers.” “This helps us conclude that both credit score and risk label are strong predictors of claim behavior.”

2. **Table:** Customer Risk Profile by Life Events

What It Shows: A breakdown of how different life events (e.g., Marriage, Job Change) are associated with risk levels.

“Marriage and job changes are the most common life events among High-Risk customers.” “Interestingly, even low-risk customers experience these events but tend to file fewer claims.” “This tells us certain life events may trigger higher-risk behavior, helping insurers flag key transition moments.”

3. **Stacked Bar Chart:** Distribution of Risk Levels Across Credit Scores

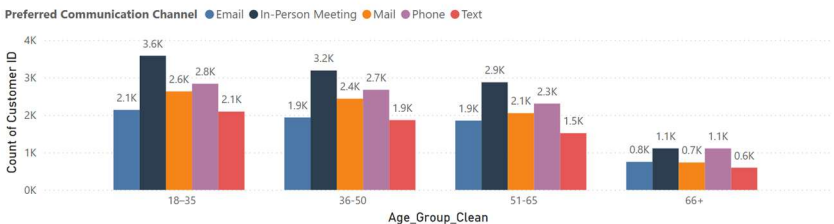
What It Shows: The proportion of risk levels (High, Medium, Low, Very Low) within each credit score group.

“In lower credit score groups (500–599), most customers fall under High or Medium risk.” “As credit scores increase, the Very Low Risk portion grows significantly—especially in the 800+ range.” “This proves a strong inverse relationship: Higher credit score leads to Lower risk level.”

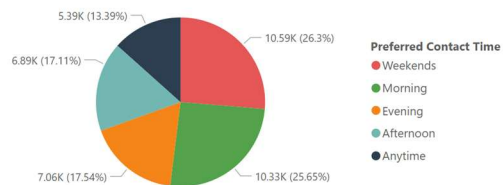
3. **Communication Preferences:**

How do customer preferences vary when it comes to communication and contact methods

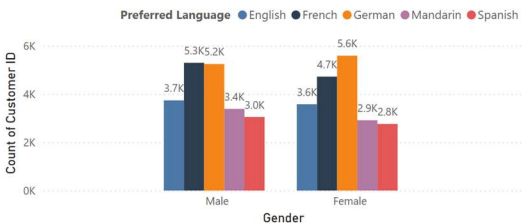
Preferred Communication Channels by Customer Segments



Preferred Contact Time by Customer Age Group



Count of Customer ID by Gender and Preferred Language



**1.Bar Chart:** Preferred Communication Channels by Customer Segments

What It Shows: Shows how different age groups prefer different communication channels

(Email, In-Person, Mail, Phone, Text).

“Customers aged 18–35 prefer email and text, while older age groups (51–65) prefer more traditional channels like mail or phone.” “Across all age groups, email is the most preferred, showing the importance of digital communication.”

**2. Pie Chart:** Preferred Contact Time by Customer Age Group

What It Shows: Displays the overall preference for contact time (Morning, Afternoon, Evening, Weekends, Anytime).

“Morning and weekend are the most preferred contact times, showing a balance between professional hours and personal convenience.” “Only a small portion (13%) prefer ‘Anytime’, meaning most customers want to be contacted at specific times.”

**3. Bar Chart:** Count of Customer ID by Gender and Preferred Language

What It Shows: Compares language preferences between males and females. “Both males and females mostly prefer English and French, but more females prefer German than males.” “This insight helps tailor communication campaigns based on both gender and language.”

**5.2 Features & Functionalities**

- Custom tooltips for detailed hover insights
- Dynamic titles and visuals based on filters
- Charts: Bar, Pie, Stacked Column, Line

6. Technical Requirements

**6.1 Software & Tools** - Microsoft Power BI Desktop: For data modeling and dashboard creation  
Microsoft Excel: Preliminary data preparation

**6.2 Hardware Requirements** - Minimum: Intel i5, 8GB RAM, SSD, Windows 10  
Recommended: Intel i7, 16GB RAM for smooth Power BI performance

7. Milestones & Timeline

| Milestone                    | Target Date  |
|------------------------------|--------------|
| Data Collection and Cleaning | June 10,2025 |
| ETL and Data Modeling        | June 12,2025 |
| Dashboard Prototype          | June 15,2025 |
| User Testing and Refinement  | June 16,2025 |
| Final Review and Submission  | June 17,2025 |



## 8. Conclusion

Insightful and practical data on consumer habits, risk mitigation strategies, and preferred methods of contact are presented in the Insurance Risk & Policy Analytics Dashboard. The scalability and usefulness of data visualization and modeling are guaranteed by the modular architecture. It is possible to refine pricing tactics or use machine learning to forecast claims in future versions. Using Power BI effectively and extracting value from structured datasets are both showcased in this project.

## Appendix:

PBIX File: policy final.pbix

Presentation: Insurance Risk & Policy Analytics Dashboard.pptx

Dataset: Kaggle – Insurance Claims and Policy Data

(<https://www.kaggle.com/datasets/ravalsmit/insurance-claims-and-policy-data>)